

EXECUTIVE INVESTMENT MEMORANDUM

PROJECT TITAN: 2-GIGAWATT AI DATA CENTER (TEXAS ERCOT REGION)

Model Architect: *Qwen 2.5 32B Instruct* | Benchmark Run: October 2024 | Classification: Tier-IV Mission Critical

This document provides an institutional investment analysis evaluating the capital deployment, civil engineering feasibility, power interconnection, and financial return profile for constructing a **2.0 Gigawatt AI hyperscale facility** in Texas. Based on the economic projections synthesized by **Qwen 2.5 32B Instruct**, total Capital Expenditure (CapEx) is modeled at **\$875,000,000** (Base Range: \$87,500,000 to \$1,750,000,000), with annualized Operational Expenditures (OpEx) of **\$40,000,000/year**. The facility is engineered to support ultra-dense AI clusters exceeding 100 kW/rack with redundant grid feeds.

Total Planned Capacity	Target Facility Area	Estimated CapEx (Mid)	Est. Annual OpEx	Target PUE
2.0 GW (2,000 MW)	100,000 sq ft	\$875,000,000	\$40,000,000	1.15 - 1.25

Capital Expenditure Allocation & Cost Centers

Item	Cost Center Description	Amount (\$USD)	Share (%)
1	Phase 1 Land Acquisition (50 Acres)	\$500,000	0.0%
2	Site Prep & Utilities	\$200,000	0.0%
3	Phase 1 Structural Buildout	\$30,000,000	1.7%
4	Electrical Wiring & Switchgear	\$10,000,000	0.6%
5	HVAC & Chiller Modules	\$10,000,000	0.6%
6	Data Center Infrastructure Servers	\$20,000,000	1.1%
7	Security & Access Control	\$1,000,000	0.1%
8	Phase 1 Contingency	\$5,000,000	0.3%
9	Full 2GW Scale Multiplier (20x Phases)	\$1,662,500,000	95.6%
TOTAL CAPITAL INVESTMENT		\$1,739,200,000	100.0%

Strategic Architecture & Critical ERCOT Grid Risk Analysis

1. ERCOT Substation Interconnection: Interconnecting a continuous 2.0 GW load requires direct 345 kV dual-circuit transmission lines, on-site utility substations, and multi-year coordination with ERCOT grid operators. Peak summer heat and grid curtailment necessitate on-site generation and battery backup.

2. Advanced Cooling Architecture: With high-density AI accelerators exceeding 1 kW per socket, standard air cooling is insufficient. The model specifies closed-loop liquid-to-liquid heat exchangers and immersion manifolds, achieving an optimized PUE while conserving scarce regional water reserves.

3. Investment Recommendation: The economics confirm a highly defensible infrastructure asset. Initial phasing in 250 MW to 500 MW data hall increments provides de-risked capital deployment and allows forward cash flows to fund sequential phases.